

The Geometric Origin of the Bekenstein–Hawking Entropy Coefficient

Why $\eta = 1/(4\ell_P^2)$ Is Not a Free Parameter in Jacobson’s Thermodynamic Derivation of Gravity

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April 2026

Zenodo DOI: [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042)

Abstract

Jacobson’s 1995 derivation of the Einstein field equations from thermodynamics makes two assumptions: entropy proportional to horizon area, $S = \eta A$, and that η is a free parameter identified with the observed Bekenstein–Hawking value $1/(4\ell_P^2)$ by matching to Newton’s constant. We show that neither assumption is independent of the causal geometry of flat spacetime.

The proportionality $S \propto A$ follows necessarily from the irreversibility of quantum-to-classical transitions (decoherence): each such transition produces classical information that must be encoded at the nearest causally accessible surface. Causal boundaries are two-dimensional — volume encoding is causally forbidden because information inside the horizon is inaccessible to external observers — so entropy scales with area, not volume. The assumption $S \propto A$ is therefore a causal necessity, not an independent postulate.

The coefficient $\eta = 1/(4\ell_P^2)$ is geometrically determined within Jacobson’s framework by the ratio of two quantities derivable from flat spacetime independently of $S \propto A$: the Einstein equation normalization 8π (from Gauss’s law in three dimensions giving 4π , doubled by the Bianchi identity trace structure) and the Euclidean Rindler cone angle 2π (the unique topological period eliminating the conical singularity at the horizon). Their ratio:

$$\frac{1}{4} = \frac{2\pi}{8\pi}$$

fixes η as a geometric consequence. The same ratio determines the minimum horizon area per decoherence event to be $4\ell_P^2$, consistent with exactly one bit per $4\ell_P^2$ — three statements of the same geometry. Jacobson’s two assumptions reduce to one derived result. The single remaining open question is why a minimum resolvable length $\ell_P = \sqrt{\hbar G/c^3}$ exists at all — a quantum gravity microstructure question that thermodynamic approaches cannot address from within their own framework.

Keywords: Bekenstein–Hawking entropy; Jacobson thermodynamic gravity; Rindler horizon; Unruh effect; holographic principle; IAM’s Law; geometric origin of Newton’s constant

1 The Problem

Jacobson ([Jacobson, 1995](#)) showed that the Einstein field equations follow from the Clausius relation $\delta Q = T dS$ applied to local Rindler horizons. The derivation requires two inputs:

1. The Unruh temperature of the Rindler horizon ([Unruh, 1976](#); [Davies, 1975](#)):

$$T = \frac{\hbar \kappa}{2\pi k_B}, \quad (1)$$

where κ is the surface gravity.

2. Entropy proportional to horizon area:

$$S = \eta A, \quad (2)$$

where η is treated as a free parameter.

Applying the Clausius relation and requiring it to hold for all null vectors yields the Einstein equations with Newton’s constant:

$$G = \frac{c^4}{4\hbar\eta}. \quad (3)$$

Jacobson then identifies $\eta = c^3/(4\hbar G) = 1/(4\ell_P^2)$ by matching Eq. (3) to the observed value of G . The coefficient 1/4 is recovered, not derived.

This gap has been shared by every thermodynamic approach to gravity since ([Cai & Kim, 2005](#); [Padmanabhan, 2010](#)). Loop quantum gravity recovers $\eta = 1/(4\ell_P^2)$ only by fixing the Barbero–Immirzi parameter ([Rovelli & Smolin, 1995](#)). String theory recovers it in specific limits ([Strominger & Vafa, 1996](#)). Neither approach derives it from the causal geometry of the horizon alone.

The present paper shows that both assumptions are geometric consequences of the causal structure of flat spacetime. Neither is independent. Section 2 derives $S \propto A$ from the irreversibility of quantum-to-classical transitions. Section 3 derives the coefficient 1/4 from the ratio $2\pi/8\pi$.

2 Why Entropy Scales with Area: The Causal Accessibility Argument

Jacobson assumed $S \propto A$ without derivation. We show it follows from three physical facts that hold independently of any entropy-area postulate.

2.1 Decoherence Produces Classical Information

Every irreversible quantum-to-classical transition — every instance of a quantum system evolving from superposition to a definite classical outcome through interaction with its environment — produces classical information. This is the decoherence process (Zurek, 2003). A quantum system $\mathcal{S}$ interacting with environment $\mathcal{E}$ evolves as:

$$|\psi\rangle_{\mathcal{S}} \otimes |E_0\rangle \longrightarrow \sum_i c_i |s_i\rangle \otimes |E_i\rangle, \quad (4)$$

where $\{|s_i\rangle\}$ are the pointer states selected by the interaction. When the environmental states $\{|E_i\rangle\}$ become orthogonal, the reduced density matrix of $\mathcal{S}$ becomes diagonal in the pointer basis:

$$\rho_{\mathcal{S}} = \text{Tr}_{\mathcal{E}} |\Psi\rangle\langle\Psi| \longrightarrow \sum_i |c_i|^2 |s_i\rangle\langle s_i|. \quad (5)$$

The system has selected a definite classical outcome. This transition is irreversible: recovering the off-diagonal elements would require erasing the produced information from the environment, which by Landauer’s principle (Landauer, 1961) dissipates $k_B T \ln 2$ per bit into the environment and increases its entropy. Each decoherence event produces $I = \log_2 N$ bits of classical information, where N is the number of distinguishable pointer states.

2.2 Classical Information Must Be Encoded at a Causally Accessible Surface

Classical information produced by a decoherence event must be permanently encoded somewhere accessible to observers outside the system. The constraint is causal: information is accessible to an external observer if and only if it lies on or outside the causal boundary of the system. Inside the horizon, no information can reach external observers by definition.

The causal boundary of a region is a two-dimensional surface — the horizon. The region itself is three-dimensional. Therefore:

- **Surface (area) encoding:** information on the 2D causal boundary is accessible to all external observers.
- **Volume encoding:** information inside the 3D bulk is causally inaccessible to external observers.

If the produced information is to be permanently accessible — as it must be, since decoherence is irreversible and the information cannot be destroyed without violating the second law — it must be encoded at the 2D causal boundary, not in the 3D bulk.

2.3 Area Scaling Is a Causal Necessity

Since every irreversible quantum-to-classical transition encodes its produced information on the 2D causal boundary, and since the information content of the boundary grows with the number of independent encoding sites on that surface, the total information encoded on the boundary scales with the boundary’s area. Therefore:

$$S \propto A. \quad (6)$$

This is not an independent postulate. It follows from: (1) decoherence produces classical information irreversibly (Eq. 4–5), (2) permanent information must be encoded at the causally accessible boundary, and (3) the boundary is a 2D surface whose information capacity scales with area.

Jacobson’s assumption $S = \eta A$ is therefore not an additional input beyond the causal geometry of spacetime. It is a consequence of the irreversibility of quantum-to-classical transitions and the causal structure of the horizon. The only remaining question is the value of η — the proportionality constant — which is derived in the following section.

3 The Two Geometric Factors Determining η

The coefficient 4 in $G = c^4/(4\hbar\eta)$ is the ratio:

$$4 = \frac{8\pi}{2\pi}. \quad (7)$$

Both 8π and 2π are geometric facts about flat spacetime. Neither assumes entropy proportional to area.

3.1 The 2π : Euclidean Rindler Cone Angle

The Rindler metric near an accelerating horizon takes the form:

$$ds^2 = -\kappa^2 \rho^2 dt^2 + d\rho^2 + d\mathbf{x}_\perp^2. \quad (8)$$

Under Euclidean continuation $t \rightarrow -i\tau$:

$$ds_E^2 = \kappa^2 \rho^2 d\tau^2 + d\rho^2 + d\mathbf{x}_\perp^2. \quad (9)$$

In the $(\rho, \kappa\tau)$ plane, this is a cone with ρ as the radial coordinate and $\kappa\tau$ as the angular coordinate. At $\rho = 0$ (the horizon), this cone has a conical singularity unless $\kappa\tau$ has period exactly 2π . This is the unique condition for a smooth, flat cone with no conical deficit:

$$\kappa\tau \sim \kappa\tau + 2\pi \implies \tau \sim \tau + \frac{2\pi}{\kappa}. \quad (10)$$

This is the KMS (Kubo–Martin–Schwinger) period of the Rindler Green’s function (Unruh, 1976; Bisognano & Wichmann, 1976). The Unruh temperature $T = \hbar\kappa/(2\pi k_B)$ follows directly: the thermal period $\beta = 1/(k_B T) = 2\pi/(\hbar\kappa)$ matches the geometric period of the Euclidean cone.

The 2π is a topological necessity. It is the full angle of the Euclidean horizon cone, fixed by the requirement that the horizon be a non-singular surface in flat spacetime. It makes no reference to η , to $S \propto A$, or to Newton’s constant.

3.2 The 8π : Einstein Equation Normalization

The Einstein field equations are:

$$G_{ab} + \Lambda g_{ab} = \frac{8\pi G}{c^4} T_{ab}. \quad (11)$$

The coefficient 8π in Eq. (11) is fixed by two independent geometric requirements:

(a) **Gauss’s law in three spatial dimensions.** In the Newtonian limit, the Einstein equations must reduce to Poisson’s equation:

$$\nabla^2 \Phi = 4\pi G \rho. \quad (12)$$

The factor 4π is the solid angle of the unit sphere in $\mathbb{R}^3$:

$$\Omega_3 = \oint d\Omega = 4\pi. \quad (13)$$

This is a purely geometric fact about three-dimensional space.

(b) **The Bianchi identity trace factor.** The Einstein tensor $G_{ab} = R_{ab} - \frac{1}{2}Rg_{ab}$ differs from the Ricci tensor by the trace term $\frac{1}{2}Rg_{ab}$. In the weak-field limit, this trace structure introduces a factor of 2 relative to the naive identification with Poisson’s equation (Wald, 1984).

Combining: $8\pi = 2 \times 4\pi$. Both factors are purely geometric. The 8π makes no reference to η , to $S \propto A$, or to the numerical value of G .

4 The Derivation

Proposition 1 (Geometric determination of the entropy coefficient). *The entropy-area proportionality constant η in Jacobson’s thermodynamic derivation of gravity is not a free parameter. It is geometrically determined by:*

$$\eta = \frac{1}{4\ell_P^2}, \quad \frac{1}{4} = \frac{2\pi}{8\pi}, \quad (14)$$

where 2π is the Euclidean Rindler cone angle and 8π is the Einstein equation normalization, both derivable from the causal structure of flat spacetime without assuming $S \propto A$.

Derivation. Jacobson’s derivation yields the relation:

$$\frac{\hbar\eta}{2\pi} = \frac{c^4}{8\pi G}. \quad (15)$$

This equation follows from applying $\delta Q = T \delta S$ to the Rindler horizon and requiring the result to hold for all null vectors. The 2π on the left enters through the Unruh temperature (Eq. 1). The 8π on the right is the Einstein normalization (Eq. 11).

Solving for η :

$$\eta = \frac{c^4}{8\pi G} \cdot \frac{2\pi}{\hbar} = \frac{c^4}{4\hbar G} = \frac{c^3}{4\hbar G/c} = \frac{1}{4} \cdot \frac{c^3}{\hbar G} = \frac{1}{4\ell_P^2}. \quad (16)$$

The coefficient $1/4 = 2\pi/8\pi$ is the ratio of the Euclidean horizon cone angle to the Einstein normalization. Both are geometric. Neither assumes $S \propto A$.

The entropy-area relation $S = \eta A$ with $\eta = 1/(4\ell_P^2)$ is therefore not a free assumption in Jacobson’s framework — it is the unique relation consistent with the causal geometry of flat spacetime. $\square$

A note on what this shows. This is not a prediction of η from outside Jacobson’s framework, nor a claim that η could have been derived without Jacobson’s machinery. It is a demonstration that η is geometrically fixed *within* that framework by the same causal geometry that determines T and G_{ab} . Jacobson had all the ingredients: the 2π from the Euclidean horizon geometry and the 8π from the Einstein normalization. Their ratio $2\pi/8\pi = 1/4$ fixes η without any observational input. The matching to the observed G is then a consistency check, not an independent determination of η .

5 Independence of the Two Factors

The derivation requires that neither factor assumes the conclusion.

The 2π from the Euclidean Rindler cone angle is a topological property of flat Minkowski spacetime near an accelerating horizon. The Euclidean continuation of the Rindler metric produces a cone in the (ρ, τ) plane. Regularity at the tip requires the angular period to be 2π — exactly the period that eliminates the conical deficit. This is a statement about flat spacetime geometry, derivable without reference to black holes, entropy, or $S \propto A$. The Bisognano–Wichmann theorem (Bisognano & Wichmann, 1976) establishes this in the full algebraic QFT setting.

The 8π from the Einstein normalization follows from requiring general relativity to reduce to Newtonian gravity in the weak-field, non-relativistic limit. The factor 4π is Gauss’s law in three spatial dimensions; the factor 2 is the trace structure of the Einstein tensor. Both are geometric properties of three-space and four-dimensional Riemannian geometry respectively. Neither references entropy, Planck’s constant, or $S \propto A$.

Independence from each other: The 2π is derived from quantum field theory in flat spacetime via the KMS condition. The 8π is derived from the differential geometry of the weak-field Einstein equations. They arise from completely different physical contexts and were identified by different historical routes — Unruh (1976) and Einstein (1915) respectively. Their ratio appearing as the coefficient of η is a structural consequence of applying thermodynamics to a spacetime with both a causal horizon and a Newtonian limit.

Uniqueness: The ratio $2\pi/8\pi = 1/4$ is not tunable. Both factors are fixed by geometric necessities of flat spacetime. The value $1/4$ is selected by the geometry, not chosen to match observation.

6 Connection to Known Results

6.1 The Minimum Encoding Area: One Bit = $4\ell_P^2$

The same geometric ratio $8\pi/2\pi = 4$ that determines η also determines the minimum horizon area disturbed by one decoherence event. This provides an independent confirmation that the three results — $S \propto A$, $\eta = 1/(4\ell_P^2)$, and the minimum encoding area — are three expressions of the same causal geometry.

The Landauer cost of one decoherence event (one bit of classical information produced) at the local horizon temperature T_H is:

$$E_{\text{bit}} = k_B T_H = \frac{\hbar \kappa}{2\pi}, \quad (17)$$

where we use $k_B T_H = \hbar \kappa / (2\pi)$ from the Unruh temperature (Eq. 1) in natural units.

This energy, deposited on the horizon, disturbs a minimum area $\delta A_{\min}$ through the gravitational coupling. The area response of a horizon to an energy perturbation is set by the Einstein equation normalization:

$$\delta A_{\min} = \frac{G}{c^4} \cdot E_{\text{bit}} \cdot 8\pi, \quad (18)$$

where the factor 8π is the Einstein normalization (Eq. 11) converting energy to area perturbation.

Substituting Eq. (17) into Eq. (18):

$$\delta A_{\min} = \frac{G}{c^4} \cdot \frac{\hbar \kappa}{2\pi} \cdot 8\pi = \frac{4G\hbar \kappa}{c^4}. \quad (19)$$

At the minimum surface gravity $\kappa_{\min} = c^2/\ell_P$ (the Planck-scale horizon, where quantum and gravitational effects are simultaneously relevant):

$$\delta A_{\min} = \frac{4G\hbar}{c^4} \cdot \frac{c^2}{\ell_P} = \frac{4G\hbar}{c^2 \ell_P} = 4\ell_P^2, \quad (20)$$

where we used $\ell_P^2 = \hbar G/c^3$ in the last step.

The minimum horizon area disturbed by one decoherence event is $4\ell_P^2$. The entropy density $\eta = 1/(4\ell_P^2)$ gives exactly one bit per $4\ell_P^2$:

$$\eta \cdot \delta A_{\min} = \frac{1}{4\ell_P^2} \cdot 4\ell_P^2 = 1 \quad (\text{bit per encoding area}). \quad (21)$$

The factor of 4 in the minimum encoding area arises from $8\pi/(2\pi) = 4$ — the same ratio that determines η . This is not a numerical coincidence. It is the same geometric statement expressed twice: the ratio of the Einstein normalization to the Euclidean horizon period determines both the entropy coefficient and the minimum encoding area, and they are consistent by construction.

6.2 Jacobson's Machine Fully Specified

Eq. (15) shows that Jacobson's derivation has the structure:

$$\underbrace{\frac{\hbar \eta}{2\pi}}_{\text{Rindler geometry}} = \underbrace{\frac{c^4}{8\pi G}}_{\text{Einstein geometry}}. \quad (22)$$

Left side: the entropy-area density scaled by the Euclidean horizon period, both set by flat spacetime geometry. Right side: the gravitational coupling set by the Newtonian limit and the trace structure of the Einstein tensor. The equation is satisfied if and only if $\eta = 1/(4\ell_P^2)$. No observational input required.

Jacobson's machine — *specify an entropy functional, derive a gravity theory* — now fully specifies its own entropy functional. The machine is self-contained.

6.3 Why Area Scaling Is Geometrically Necessary

The $S \propto A$ relation is not merely consistent with the causal geometry — it is the unique scaling required by it. Information from bulk decoherence events can only be permanently encoded on the causal boundary: anything inside the horizon is causally disconnected from external observers. Bulk volume encoding would place information inaccessible to those observers. Area scaling is therefore a causal necessity ('t Hooft, 1993; Susskind, 1995), and the coefficient $1/(4\ell_P^2)$ is the unique value consistent with the Rindler and Einstein geometric factors derived above.

6.4 The Same Projection Structure at Cosmological Scale

The ratio $2\pi/8\pi = 1/4$ involves the same geometric projection factors — a full-sphere solid angle (4π), its doubling (8π), and a topological period (2π) — that appear in the IAM derivation of the cosmological constant (IAM archive, 2026). In that context, the de Sitter causal patch factor $2/\pi$ and its inverse $\pi/2$ (the reverse holographic projection) appear independently in the CC derivation and in the empirical Koide formula for lepton masses (IAM archive, 2026; Koide, 1983), with their product being exactly 1. The present result identifies the Planck-scale analog: the ratio $2\pi/8\pi$ at the horizon scale corresponds to the same holographic projection structure operating at the fundamental causal boundary of flat spacetime.

7 The Remaining Open Problem

The present results close three questions that Jacobson left open:

(A) Why does S scale with A ? Derived in Section 2: decoherence produces classical information irreversibly; permanent information must be encoded at the causally accessible 2D boundary; volume encoding is causally forbidden. Area scaling is a necessary consequence of the causal structure.

(B) Why is the coefficient $1/4$? Derived in Section 3: $1/4 = 2\pi/8\pi$, where 2π is the Euclidean Rindler cone angle and 8π is the Einstein normalization. Both are geometric facts about flat spacetime. Neither assumes $S \propto A$.

(C1) Why does one bit occupy $4\ell_P^2$? Derived in Section 6: the minimum horizon area disturbed by one Landauer cost event at Planck-scale surface gravity is $4\ell_P^2$, from the same $8\pi/2\pi = 4$ ratio. The entropy coefficient and the minimum encoding area are the same geometric fact.

The remaining open question is distinct from all three above. It is:

Why does a minimum resolvable length $\ell_P = \sqrt{\hbar G/c^3}$ exist at all?

$\ell_P^2 = \hbar G/c^3$ is the unique area constructible from the three constants governing quantum mechanics ($\hbar$), gravity (G), and relativity (c). If a minimum encoding area exists, it must be proportional to ℓ_P^2 by dimensional necessity — no other combination is available. But why a minimum area exists in the first place, and why the three constants $\hbar$, G , c combine to set it, is a question about the discrete microstructure of spacetime at the Planck scale. This lies beyond the thermodynamic level of Jacobson's derivation and cannot be addressed from within it.

Loop quantum gravity and string theory approach this question from different directions. The present derivation is neutral between those approaches. It identifies ℓ_P^2 as the minimum

unit by dimensional necessity, and derives all three geometric factors ($S \propto A$, $\eta = 1/4$, one bit per $4\ell_P^2$) from causal spacetime structure. The single remaining gap is the quantum gravity account of why Planck-scale discreteness exists.

8 Discussion

The result is best stated precisely rather than broadly.

Jacobson’s 1995 paper made two assumptions: $S \propto A$ and η is a free parameter. Both were reasonable at the time. $S \propto A$ was motivated by the Bekenstein–Hawking result for black holes; η was recovered by matching to Newton’s constant. What was not recognized is that both assumptions follow from the causal geometry of flat spacetime.

$S \propto A$ follows from the irreversibility of decoherence and the causal inaccessibility of the bulk: information produced by an irreversible quantum-to-classical transition must be encoded at the 2D causal boundary because nowhere else is causally accessible to external observers. This does not require quantum gravity; it requires only the causal structure of spacetime and the irreversibility of decoherence established by Landauer’s principle (Landauer, 1961) and Zurek’s decoherence framework (Zurek, 2003).

$\eta = 1/(4\ell_P^2)$ follows from the ratio $2\pi/8\pi$ where both factors are present in Jacobson’s derivation: the 2π from the Euclidean Rindler cone angle (Bisognano–Wichmann (Bisognano & Wichmann, 1976)), and the 8π from the Einstein normalization (Wald (Wald, 1984)). Jacobson had both. He did not divide them.

The minimum encoding area $4\ell_P^2 = \delta A_{\min}$ follows from the Landauer cost of one decoherence event at Planck-scale surface gravity, with the same $8\pi/2\pi = 4$ geometric ratio. The product $\eta \cdot \delta A_{\min} = 1$ is exact — one bit per minimum encoding area — and is not a separate result but the same result stated as a consistency check.

Three things Jacobson assumed are now derived. One thing remains open: why Planck-scale discreteness exists at all. That is the genuine quantum gravity question, and it is a much narrower and more precisely stated gap than any existing treatment of the Bekenstein–Hawking coefficient provides.

The thermodynamic level of the derivation of $S = A/(4\ell_P^2)$ is complete.

Acknowledgments

The thermodynamic framework of T. Jacobson (Jacobson, 1995) is the foundation of this work. This paper was motivated directly by a personal response from Professor P.J.E. Peebles, who wrote: *“It is good to think independently, as you have done by exploring consequences of the entanglement of the whole world, assuming quantum physics is accurate enough to be trusted for this purpose.”* That statement — identifying the quantum-to-classical derivation as the foundational gap requiring closure before the cosmological framework could be trusted — directly motivated the analysis presented here. The Unruh–Rindler geometry of W.G. Unruh (Unruh, 1976) and the parallel result of P.C.W. Davies (Davies, 1975) provide the 2π factor. The Bisognano–Wichmann theorem (Bisognano & Wichmann, 1976) establishes it in the full QFT setting.

Data Availability

All derivations, numerical verifications, and companion papers are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> with canonical Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042).

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